

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

I. Omar and Trees

time limit per test: 1 second
memory limit per test: 256 megabytes

When Omar was at school, his teacher gave him a challenge, and Omar accepted it.

The teacher gave Omar a tree of n nodes (node 1 is the root). The i_{th} node has a value a_i .

Then the teacher asked Omar q queries of two types:

- Update the value in every node in the subtree rooted with u to val (u included).
- Determine if the sum of all nodes in the subtree with root u can be represented as the sum of two prime numbers or not (u included).

Omar asks for your help in this challenge.

Input

The first line contains one integer n ($1 \leq n \leq 10^5$).

The second line contains n integers ($1 \leq a_i \leq 10^5$)—the value of each node.

For each $n - 1$ lines, it contains two integers u and v ($1 \leq u, v \leq n, u \neq v$)— indices of nodes connected by an edge.

Then the number of queries q ($1 \leq q \leq 10^5$).

Each query format is one of the following:

- 1 u val : Update the value in every node in the subtree rooted with u to val ($1 \leq val \leq 10^5$).
- 2 u : Determine if the sum of all nodes in the subtree with root u can be represented as the sum of two prime numbers or not.

Output

For each query of the second type, print "YES" without quotes if the sum of all nodes in the subtree with root u can be represented as the sum of two prime numbers or not, "NO" without quotes otherwise.

Example

input	Copy
9 3 5 2 7 10 6 1 4 3 1 2 2 3 1 4 4 5 5 6 4 7 7 8 7 9 6 2 7 2 2 1 7 10 2 7 1 7 9 2 7	
output	Copy
YES YES YES NO	

TCPC Tunisian Collegiate Programming Contest 2022

Finished

Practice

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Last submissions

Submission	Time	Verdict
362703543	Feb/13/2026 00:59	Accepted
320697701	May/21/2025 23:45	Accepted
320691426	May/21/2025 22:03	Accepted
320689556	May/21/2025 21:39	Accepted
317890399	May/01/2025 04:52	Accepted
317683350	Apr/29/2025 03:30	Accepted
317683172	Apr/29/2025 03:23	Wrong answer on test 2
317528948	Apr/28/2025 06:22	Accepted
317511131	Apr/27/2025 23:28	Accepted
317511105	Apr/27/2025 23:27	Wrong answer on test 5